ARS POLICY BRIEF

Beyond the Car Factory: Where Pakistan's EV Opportunity Really Lies

Market intelligence for Chinese investors, manufacturers and technology partners

By Reza Khan

29 August 2026

Central investment thesis

Pakistan's EV opportunity may be larger for Chinese investors in batteries, components, charging, commercial mobility and two and three-wheelers than in conventional passenger-vehicle assembly alone. This report tests that proposition against the available evidence and concludes that it holds, with qualifications. The volume, the unit economics and the policy support all sit in the small-vehicle and energy-storage layers of the value chain, while the passenger-car segment that attracts most attention remains a market of a few thousand units a year competing for scarce foreign exchange.

Reference: ARS-PB-2026-001

Suggested citation: Khan, R. (2026). Beyond the Car Factory: Where Pakistan's EV Opportunity Really Lies. Ardent Research Solutions. www.arspakistan.pk

All figures, scores and rankings marked ARS are the analytical product of Ardent Research Solutions and are set out with their methodology in the final section of this report.

1. Executive Summary

Pakistan is electrifying from the bottom of the vehicle pyramid upward, and the investment case for Chinese companies follows that shape rather than the shape of the passenger-car market. In fiscal year 2026 the country sold approximately 1.93 million motorcycles and 42,000 three-wheelers, an all-time record, while the entire market for electric and plug-in hybrid passenger cars was measured in low thousands of units. Electric two-wheeler sales grew by more than 200% year on year in the first half of 2026, placing Pakistan among the five largest electric two-wheeler markets in the world by volume growth. The scale, the fuel-price arbitrage and the state subsidy are all concentrated in the segment that Chinese manufacturers already dominate globally.

The New Energy Vehicle Policy 2025-30, approved by the federal cabinet on 26 August 2025, targets 30% of new vehicle sales as new energy vehicles by 2030, equivalent to an implied 2.2 million vehicles, rising to 90% by 2040, alongside 3,000 charging stations. The policy is enacted and funded to a degree, through the NEV Adoption Levy Act 2025 and the Pakistan Accelerated Vehicle Electrification programme, but the delivery record is uneven. Phase 1 of that programme targeted 41,000 vehicles and had delivered approximately 5,409 by the time the Economic Coordination Committee reviewed it in May 2026, against roughly 269,149 applications. The binding constraint was bank-side credit processing rather than consumer demand, which is a fixable problem and a commercially significant one for any investor considering a captive financing arm.

For Chinese investors the most attractive positions are upstream and adjacent to vehicles rather than in vehicle assembly itself. Battery-pack assembly for two and three-wheelers is the strongest single opportunity, because Pakistan imported an estimated 1.25 GWh of lithium-ion packs in 2024 and faces demand of roughly 8.7 GWh by 2030 on official projections, while the entire domestic capacity announced to date amounts to a handful of Karachi plants with output measured in thousands of units per month. Battery swapping, charging networks and non-electric component supply follow. Passenger EV assembly, the segment attracting the most publicity through the 150 million dollar BYD and Mega Motor plant at Gharo, is in ARS's assessment the least attractive point of entry for a new investor, because the addressable market is thin, the incumbent is vertically integrated and the capital intensity is high.

The risks are concentrated in policy continuity and foreign exchange rather than in demand. Pakistan has changed automotive tariff policy repeatedly since 2020, the Automotive Industry Development and Export Plan expires in 2026, and the successor Auto Policy 2026-31 and National Lithium-Ion Battery Manufacturing Policy 2026-31 were both still pending approval at the time of writing. Foreign-exchange conditions have improved materially, with 2.305 billion dollars in profits and dividends repatriated in fiscal year 2026 and no reported official backlog, but that improvement is recent, programme-dependent and reversible.

ARS Key Findings

1. The mass market is electric two and three-wheelers, not cars. Approximately 160,000 electric motorcycles have been produced locally against roughly 12,800 electric four-wheelers, and more than 60 manufacturers hold licences in the small-vehicle categories.

2. Battery-pack assembly is the highest-scoring opportunity in the ARS matrix at 78 out of 100, because import dependence is near total, the customer base is fragmented and domestic, and minimum efficient scale is low.

3. Charging infrastructure is undersupplied but not yet independently bankable. The subsidised station tariff of approximately 39.70 rupees per kilowatt-hour improves operating economics, but utilisation rather than tariff is the binding constraint at present vehicle densities.

4. The tariff structure supports assembly more than it supports localisation. Zero-rated plant, machinery and vendor inputs sit alongside a 1% duty on completely knocked-down kits, which keeps kit importing rational for longer than the stated localisation objective implies.

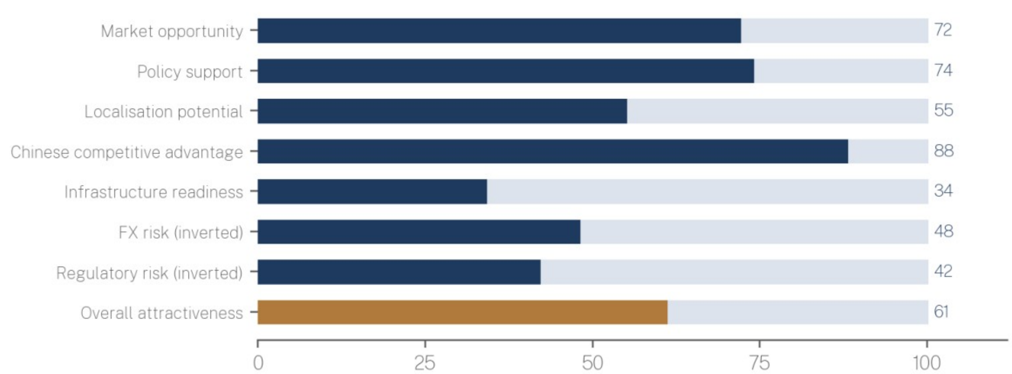
5. The 2030 target requires roughly 82% compound annual growth in the new energy vehicle fleet from an estimated base of 200,000 vehicles. ARS assesses the target as highly challenging rather than achievable on current policy settings.

6. Foreign-exchange risk has moved from acute to manageable, but component-heavy business models remain the most exposed, because they combine continuous import payments with rupee revenue and dollar-denominated returns.

7. Pakistan ranks fifth of six in the ARS China+1 comparison at 54 out of 100, behind Vietnam, Thailand, Indonesia and India. Its case rests on domestic demand and Chinese incumbency, not on export platform economics.

2. ARS EV Investment Dashboard

Figure 8 | ARS EV Investment Dashboard: Pakistan, August 2026



Scores are 0 to 100, higher is better. Risk dimensions are inverted so that a higher score denotes lower risk. Overall attractiveness is the weighted composite described in the methodology note.

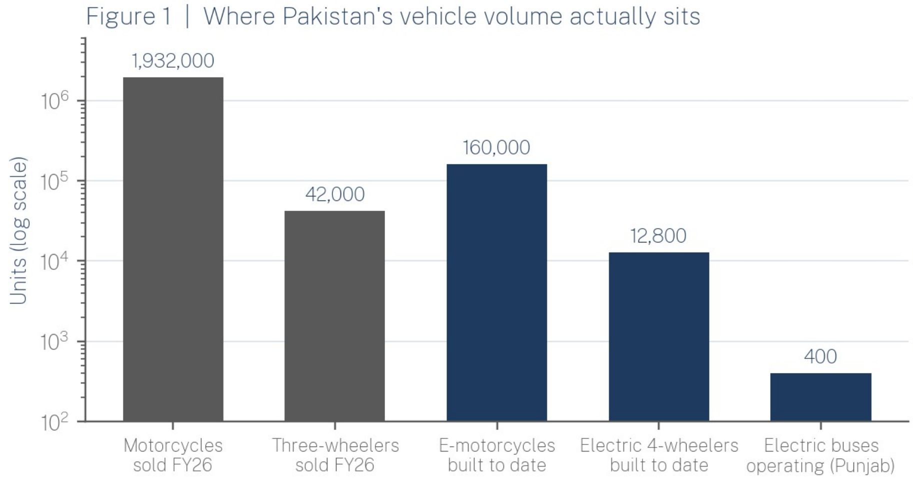
The dashboard scores Pakistan on eight dimensions on a scale of 0 to 100, with risk dimensions inverted so that a higher score always denotes a more favourable condition. Market opportunity scores 72 on the strength of two and three-wheeler volume and fuel-price arbitrage. Policy support scores 74 because incentives are legislated, funded and applied across the whole vehicle range rather than confined to passenger cars. Chinese competitive advantage scores 88, the highest of the eight, reflecting near-total Chinese dominance of the technologies Pakistan is buying, from cells and controllers to swapping cabinets and electric buses. Infrastructure readiness scores 34, the lowest, reflecting a public charging network still counted in tens rather than thousands of points and a grid whose reliability varies by city. The composite attractiveness score of 61 places Pakistan in the selective-entry band: attractive for targeted, capital-light positions in the small-vehicle and battery layers, and not yet attractive for capital-intensive integrated manufacturing.

3. Market Baseline: What the Numbers Actually Show

Pakistan's electric vehicle data is fragmented across at least four bodies, and the figures in public circulation are frequently not comparable. Officials briefing the Senate Standing Committee on Industries and Production in 2026 stated that more than 12,800 electric vehicles and nearly 160,000 electric motorcycles had been manufactured in Pakistan, while separate reporting in mid-2026 referred to more than 80,000 electric vehicles operating on Pakistani roads and more than 60 licensed manufacturers of electric motorcycles and rickshaws. These are different measures. Cumulative manufacture is not the same as registration, registration is not the same as vehicles in operation, and neither captures imported completely built-up units.

The reliable anchor is the internal combustion market. Motorcycle sales reached approximately 1.932 million units in fiscal year 2026, the highest in Pakistan's history, exceeding the previous peak of 1.853 million recorded in fiscal year 2021, with three-wheelers adding roughly 42,000 units. Atlas Honda alone accounted for approximately 1.69 million units, or close to 88% of the two-wheeler market. Against that base, an installed electric two-wheeler population of roughly 160,000 units represents penetration of well under 1% of the operating fleet, but electric two-wheeler sales grew by more than 200% year on year in the first half of 2026, and two domestic brands, Evee and Okla, recorded growth of 410% and 276% respectively over the same period.

Figure 1 | Where Pakistan's vehicle volume actually sits



Sources: Arif Habib Limited (FY26 sales); Senate Standing Committee on Industries briefing, 2026; Punjab Transport Department. Sales and cumulative production are different measures and are not additive.

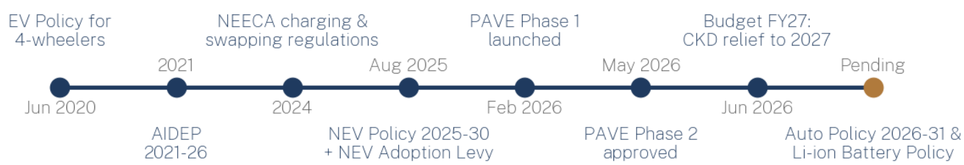
The passenger EV picture is much smaller. The combined market for electric and plug-in hybrid cars was approximately 1,000 units in 2024, and BYD is estimated by local media to have sold approximately 2,000 units in 2025, a figure the company has not confirmed. ARS's assessment is that public EV statistics in Pakistan should be treated as directionally reliable and numerically imprecise, and that any investment model should be built on the two-wheeler sales series and the subsidy quota rather than on aggregate EV counts.

4. The Policy Framework: Enacted, Funded and Pending

The New Energy Vehicle Policy 2025-30 was developed by the Ministry of Industries and Production through a steering committee convened in September 2024 and approved by the federal cabinet on 26 August 2025. Its headline commitments are 30% of new vehicle sales as new energy vehicles by 2030, equivalent to an implied 2.2 million vehicles, 90% by 2040, 3,000 charging stations nationwide, and 40 charging points on motorways at average intervals of 105 kilometres. Government projections attached to the policy include annual fuel savings of 2.07 billion litres and approximately 1 billion dollars in foreign-exchange savings. These are policy targets and official projections, not forecasts, and this report treats them as statements of intent.

What is enacted and funded is narrower but more useful to investors. The NEV Adoption Levy Act 2025 imposes a levy of 1% to 3% on conventional vehicles and funds the Pakistan Accelerated Vehicle Electrification programme on a revenue-neutral basis. The Economic Coordination Committee approved a 9 billion rupee package for fiscal year 2025-26 covering 116,053 electric bikes and 3,171 electric rickshaws and loaders, with 25% of the subsidy reserved for women. Phase 2, approved on 5 May 2026, retained a 9 billion rupee envelope and a target of 119,170 vehicles, replaced balloting with first-come-first-served allocation, and moved the subsidy to point of sale so that the Engineering Development Board reimburses the manufacturer rather than the buyer. The per-unit subsidy is 80,000 rupees for an electric motorcycle and up to 400,000 rupees for a rickshaw or loader.

Figure 4 | Pakistan EV policy timeline: enacted measures in navy, pending in amber

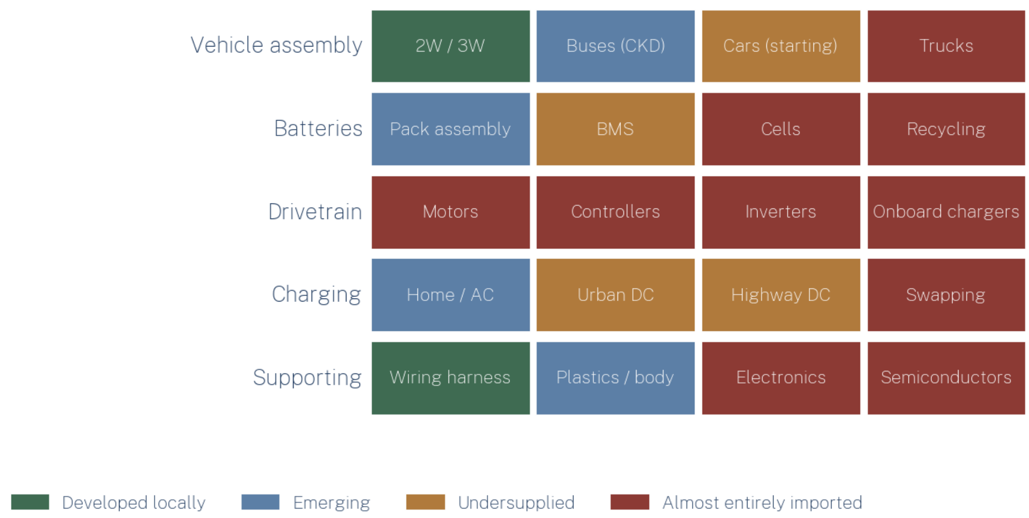


Two significant instruments remain proposals rather than policy. The Auto Policy 2026-31, successor to the Automotive Industry Development and Export Plan 2021-26, was under review by the Prime Minister's Committee at the time of the fiscal year 2026-27 budget and had not been laid before Parliament. The National Lithium-Ion Battery Manufacturing Policy 2026-31 had been forwarded by the Engineering Development Board to the Ministry of Industries and Production and was awaiting consideration by the National Tariff Board on the question of duty reductions for battery components. Both will materially affect investment returns in the segments this report identifies as most attractive, and neither should be modelled as certain.

5. Mapping the Value Chain

Pakistan's EV value chain is developed at the two ends and empty in the middle. Two and three-wheeler assembly is genuinely local, with more than 90% of parts for those categories reportedly manufactured domestically and more than 118 assemblers licensed across the two, three and four-wheeler categories. Wiring harnesses, plastics, seats and body components are supplied by an existing automotive vendor base built over three decades of internal combustion assembly. At the other end, charging hardware and vehicles can be imported readily. What does not exist is the electrochemical and power-electronics layer between them.

Figure 2 | ARS EV value-chain investment map



The commercial implication is that the localisation opportunity is not evenly distributed. Segments shaded as almost entirely imported, including cells, motors, controllers, inverters and onboard chargers, are the segments where Chinese suppliers hold the strongest technological position and where Pakistan has the least capacity to substitute. For most of these, continued import combined with local assembly is the economically rational structure for the next 3 to 5 years, and the investable position is distribution and after-sales rather than manufacturing.

6. Opportunity 1: Battery Packs, Cells and Storage

Battery-pack assembly is the strongest single opportunity identified in this report. Pakistan imported an estimated 1.25 GWh of lithium-ion battery packs in 2024, and demand is projected to reach approximately 8.7 GWh by 2030 in line with the NEV Policy. ARS notes a substantial discrepancy in official demand estimates: the Engineering Development Board has separately cited demand of 65 to 100 gigawatts by 2030, a figure that conflates power with energy and is an order of magnitude above the 8.7 GWh projection. ARS regards the 8.7 GWh figure as the more defensible planning number and treats the larger estimate as an aspiration attached to an export ambition rather than a demand forecast.

Domestic capacity is at an early stage and is mostly pack assembly rather than cell manufacture. EV Technologies is commissioning a plant in Korangi, Karachi with an initial capacity of 4 megawatts, sufficient for approximately 2,000 electric bikes and scooters per month, using nickel manganese cobalt chemistry. Crown Group is completing a lithium iron phosphate plant at Port Qasim projected at 5,000 to 8,000 batteries per month, Wave Tech has announced a 200 million dollar plant at Malir Industrial Park, and Zilo Energy has installed a lithium iron phosphate assembly line at Korangi for stationary storage. A Chinese company has committed 15 million dollars to a lithium battery plant in a special economic zone near Faisalabad under an agreement with the Punjab Board of Investment and Trade, and the Saigol Group signed a memorandum of understanding with China's Juhang Energy Technology

Group in mid-2026. These are announcements and commitments at varying stages, and only the smallest of them is near production.

Does Pakistan have the scale to justify local battery-pack assembly? For two and three-wheeler packs, yes. A subsidy quota of approximately 119,000 vehicles a year, a two-wheeler market approaching 2 million units and a rapidly growing solar storage market together support pack assembly at a scale of hundreds of megawatt-hours, and pack assembly has low minimum efficient scale, modest capital requirements and high logistics costs when imported, which is the classic profile for localisation. For cells, no. Cell manufacture requires demand an order of magnitude larger, uninterrupted high-quality power and a supply chain for cathode and anode materials that Pakistan does not have. The realistic structure is a joint venture assembling imported Chinese cells into locally made packs with locally integrated battery management systems, sold into both the vehicle and the solar storage markets, with lithium iron phosphate chemistry favoured for its thermal tolerance in Pakistani ambient conditions and its lower cost per cycle.

7. Opportunity 2: Battery Swapping

Battery swapping in Pakistan is currently a set of pilots rather than an industry, which is precisely what makes it an investment opportunity rather than a service line. The most advanced public deployment is in Faisalabad, where the district administration inaugurated the first public-sector swapping station in August 2026 with two cabinets, 24 charging slots and five electric bikes, and set a target of 50 cabinets across the city by December 2026. A donor-supported Battery Swapping Network project for electric three-wheelers across Punjab was approved for implementation in mid-2026 with a five-year horizon. Private operators including ezBike run swapping services for two-wheelers, and Neubolt has piloted three-wheeler swapping with an oil marketing company.

The economics favour high-utilisation commercial users rather than private commuters. A rickshaw or loader operating 10 to 14 hours a day cannot absorb multi-hour charging downtime, and battery cost represents the largest single component of vehicle price, so separating battery ownership from vehicle ownership reduces the purchase price barrier that the subsidy is currently addressing with public money. The addressable fleet is large, with the three-wheeler population estimated at approximately 2 million vehicles. The principal barrier is standardisation. Pakistan has no mandated swappable-battery standard, manufacturers use incompatible form factors, and a network built around one manufacturer's pack is a captive network. For a Chinese investor with an established cabinet and pack standard, the strategic prize is setting that standard early, before regulation does it for them or fragmentation makes it uneconomic.

8. Opportunity 3: Charging Infrastructure

Pakistan's public charging network is small and corridor-concentrated. HUBCO Green has installed 19 public direct-current fast-charging stations along the 1,300 kilometre Karachi to Peshawar corridor, and public estimates of total public charging points nationwide have remained in the range of tens rather than hundreds. The Sindh government has approved more than 600 stations for key cities and highways, and Punjab allocated 40 billion rupees in the fiscal year 2026-27 budget for charging infrastructure to support its electric bus fleet. The regulatory framework is now reasonably clear: the National Energy Efficiency and Conservation Authority regulates charging and swapping installations under regulations first issued in 2024, a one-window approval mechanism has been introduced, and the Power Division reduced the applicable electricity tariff for certified stations from approximately 71 rupees to approximately 39.70 rupees per unit, with charging stations also exempted from monthly fuel charge adjustments.

That tariff treatment is a genuine and unusually well-designed incentive. It is not, however, sufficient to make standalone public charging bankable at current vehicle densities, because the constraint is utilisation. With a passenger EV population in the low tens of thousands nationally and most private owners charging at home overnight, a merchant fast-charging network cannot reach the throughput that justifies its capital cost. The models that work in Pakistan today are anchored charging rather than merchant charging: fleet depot charging for electric buses, where the customer is a provincial government with a committed fleet, and charging attached to an existing footprint of fuel retail, hospitality or utility assets. ARS's assessment is that the charge-point operator model is premature as a standalone business and attractive as a bundled position for an investor who is also selling vehicles, batteries or power.

9. Opportunity 4: Motors, Controllers and Power Electronics

Electric motors, motor controllers, inverters, direct-current converters and onboard chargers are almost entirely imported, overwhelmingly from China, and there is no meaningful domestic production of any of them. Pakistan has relevant adjacent capability in electrical equipment and motorcycle vending, but lacks the precision magnet supply chain, the semiconductor access and the test infrastructure that motor and controller manufacture requires.

The realistic near-term localisation candidates are the least technology-intensive items in this group: wiring harnesses and connectors, which are already made locally, low-voltage controllers for two-wheelers, and battery management system integration and firmware, which is a software and testing activity rather than a fabrication one. Motors, inverters and power semiconductors should be assumed to remain imported for 5 years or more. For a Chinese component supplier, the commercially sensible position in this layer is a technical partnership with an existing Pakistani vendor supplying the licensed two-wheeler assemblers, rather than a greenfield plant.

10. Buses, Commercial Vehicles and the Public Procurement Channel

Commercial electrification in Pakistan is being driven by provincial procurement rather than by private fleet economics, and that makes it a different kind of opportunity, with a different risk profile. Punjab has approximately 400 electric buses operating across 18 districts, has approved procurement of 1,500 buses for 91 tehsils in a first phase, is procuring 1,100 buses at a reported cost of 94 billion rupees, and has stated an intention to operate 5,000 electric buses within 5 years. Khyber Pakhtunkhwa is adding 52 Chinese-built buses to the Peshawar Bus Rapid Transit fleet, taking it to 296 vehicles. Sindh has approved 250 diesel-hybrid buses under a separate intra-district programme.

Total cost of ownership favours electric buses in this application even at Pakistani electricity prices, because urban bus duty cycles are high-mileage, predictable and depot-based, which maximises the fuel-cost advantage and minimises range risk. The commercial risk is not technical but fiscal and political: the customer is a provincial government, payment terms follow provincial budget cycles, and procurement decisions are subject to electoral change. For Chinese bus manufacturers already supplying these fleets, the higher-margin follow-on opportunity is depot charging infrastructure, spare parts, battery replacement and maintenance contracts, which convert a one-time export sale into a recurring rupee revenue stream. Electric trucks, by contrast, carry a 1% sales tax in completely built-up condition but have almost no demonstrated demand, and ARS regards that segment as a watch item rather than an opportunity.

11. Two and Three-Wheelers: Pakistan's Actual EV Mass Market

Every quantitative indicator in this report points to the same conclusion: Pakistan's EV transition is a two and three-wheeler transition. The country has more than 26 million motorcycles on the road consuming roughly 3 billion dollars of fuel a year, sells close to 2 million new units annually, and produces more than 90% of two and three-wheeler components domestically. A typical electric two-wheeler costs approximately 250,000 rupees against approximately 160,000 rupees for the best-selling internal combustion commuter motorcycle, a gap of roughly 90,000 rupees that the 80,000 rupee federal subsidy very nearly closes on its own, before accounting for running-cost savings against petrol at current prices.

Provincial schemes reinforce the federal one. Punjab has committed to providing 100,000 electric bikes to students within a year with a subsidy of 70,000 rupees per unit, a down payment of 14,000 rupees for male students and interest-free monthly instalments of 2,100 rupees, with the province covering the down payment and registration for female students. This is a structurally subsidised, financing-led market, and that is the critical commercial insight: the constraint on volume has been credit processing, not price or demand. Phase 1 of the federal programme generated approximately 269,149 applications against 41,000 places and delivered approximately 5,409 units, a failure of bank-side approval rather than of consumer appetite.

For Chinese participants this creates four distinct entry points. Manufacturers can supply completely knocked-down kits into the licensed local assembler base, of which more than 60 exist in the small-vehicle categories, without building a plant. Battery and swapping companies can supply the energy layer, which is the highest-value and least localised part of the vehicle. Fleet and ride-hailing operators can aggregate demand for delivery and last-mile logistics, where duty cycles justify swapping. Financing partners can address the bottleneck that has actually constrained the market, and a captive non-bank financing vehicle attached to a manufacturer is, in ARS's assessment, one of the least contested and most defensible positions available in Pakistani electric mobility today.

12. Localisation: What Pakistan Can Realistically Manufacture

Component	Import dependence	Local capability	Capital required	Localisation horizon	Chinese opportunity
Wiring harnesses, connectors	Low	Established	Low	Already local	Technical upgrade, JV
Body, plastics, seats, frames	Low	Established	Low	Already local	Supply to assemblers
Battery packs (2W and 3W)	Near total	Emerging	Low to medium	0 to 2 years	Strong: pack JV
Battery management systems	Near total	Weak but feasible	Low	2 to 3 years	Strong: licence and integrate
Chargers, AC units and cabinets	Near total	Weak	Low	2 to 3 years	Moderate: assembly
Low-voltage controllers (2W)	Total	Weak	Medium	3 to 5 years	Moderate: partnership
Electric motors	Total	Very weak	High	5 years or more	Weak: continue import

Component	Import dependence	Local capability	Capital required	Localisation horizon	Chinese opportunity
Inverters and power electronics	Total	Very weak	High	5 years or more	Weak: continue import
Battery cells	Total	None	Very high	5 years or more	Weak until demand scales
Semiconductors	Total	None	Prohibitive	Not realistic	None

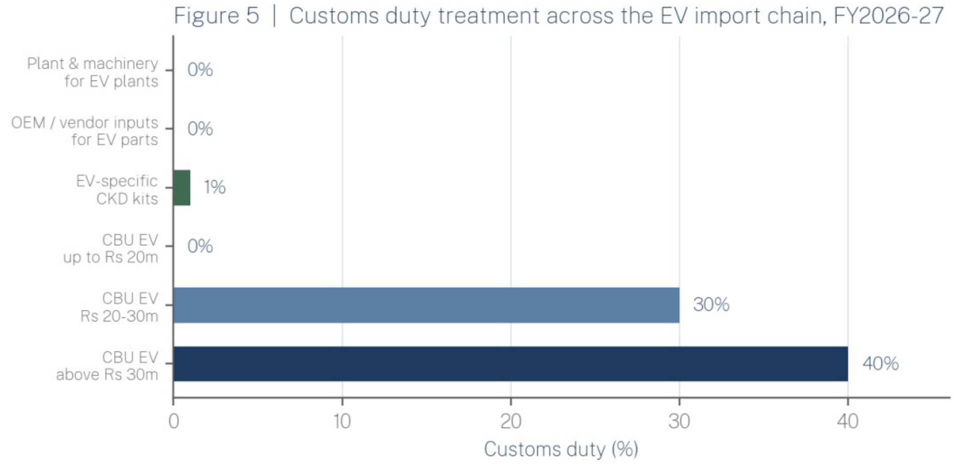
ARS assessment. Horizons denote when domestic production could plausibly be commercially competitive, not when it could technically begin.

The pattern is consistent. Where Pakistan already manufactures, it does so competitively and Chinese firms should buy rather than build. Where the technology is electrochemical or power-electronic, importing Chinese components and performing final assembly in Pakistan will remain the economically rational structure for the foreseeable future, and any policy that attempts to force localisation of those items faster than demand supports will raise costs rather than build capability. The one category that is genuinely poised to move is battery packs, which is why it dominates the ARS opportunity ranking.

13. Tariffs, Taxes and the Localisation Contradiction

The fiscal year 2026-27 budget preserved the concessional regime for electric vehicles rather than withdrawing it, contrary to widespread pre-budget expectation that the 1% sales tax on locally manufactured electric vehicles would rise to the standard 18%. Concessions on completely knocked-down kits and EV-specific parts were extended to 30 June 2027, with associated customs concessions valued at approximately 23.86 billion rupees. The 1% sales tax on locally manufactured electric vehicles and the 8.5% rate on locally manufactured hybrids up to 1800cc were retained. Plant and machinery imported for EV manufacturing facilities attracts zero customs duty on a one-time basis for new projects and expansions subject to Engineering Development Board approval, and vendor and original equipment manufacturer inputs used to produce EV-specific parts are likewise zero-rated.

Figure 5 | Customs duty treatment across the EV import chain, FY2026-27



Source: Finance Bill 2026 as reported. ARS notes an unresolved discrepancy in press reporting on completely built-up EVs, with some accounts describing a continuing 25% duty alongside the value-banded structure shown here.

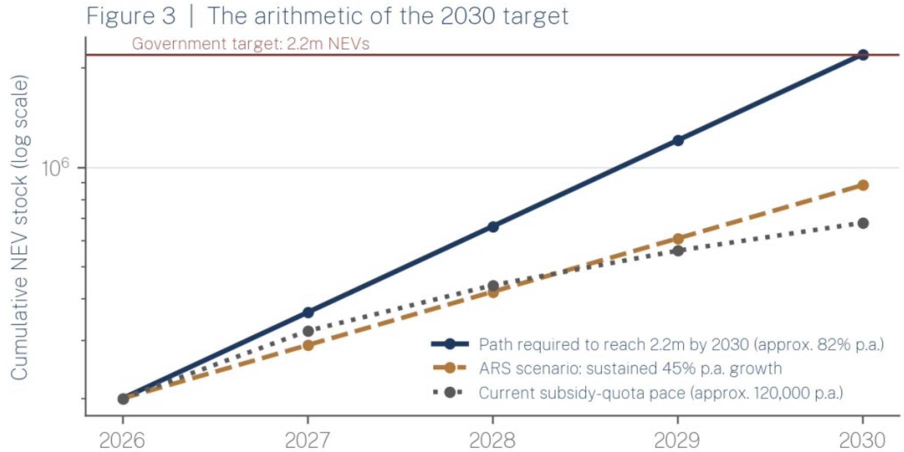
ARS flags a data-integrity issue that investors should resolve directly with the Federal Board of Revenue before modelling. Reporting on the Finance Bill 2026 describes a value-banded structure for completely built-up electric vehicles imported for personal use, with zero duty up to 20 million rupees, 30% between 20 and 30 million rupees and 40% above 30 million rupees, alongside new federal excise duty on vehicles above 20 million rupees. Other reporting from the same period states that completely built-up electric vehicles continue to attract a 25% import duty. These accounts are not reconcilable, and the practical rate applicable to a given import will depend on notification detail rather than on budget-speech summaries.

The policy-intelligence finding is this: the stated objective of localisation conflicts with the actual tariff structure. A 1% duty on completely knocked-down kits, combined with zero-rated vendor inputs and zero-rated machinery, makes importing a nearly complete vehicle almost as cheap as making one, and provides no escalating pressure to deepen local content over time. The Automotive Industry Development and Export Plan tariff facility is scheduled to continue to 2026 and be phased out gradually by 2030, but the phase-out schedule sits in a policy that has not yet been enacted. For an investor, the correct reading is that the current structure rewards assembly speed rather than localisation depth, that this favours early entrants with kit supply chains already in place, and that the risk sits with anyone who invests in deep localisation on the assumption that tariff escalation will eventually protect it.

14. The 2030 Question, Policy Continuity and Foreign Exchange

On current evidence, ARS assesses Pakistan's 2030 new energy vehicle target as highly challenging rather than achievable. Reaching an implied stock of 2.2 million new energy vehicles by 2030 from an estimated base of approximately 200,000 in mid-2026 requires compound annual growth of roughly 82%, sustained for four consecutive years. Electric two-wheeler sales have grown faster than that in individual recent periods, which is the strongest argument for the target, but that growth is subsidy-driven and the subsidy envelope is quota-limited at approximately 119,000 vehicles a year. Sustaining 82% growth would require either a large expansion of the levy-funded subsidy pool or a shift to unsubsidised price parity, and neither is currently scheduled.

Figure 3 | The arithmetic of the 2030 target



ARS calculation. Baseline of approximately 200,000 NEVs in mid-2026 is an ARS estimate reconciling official production and operating-fleet figures; the government target is a sales-share target expressed as an implied stock of 2.2m vehicles.

Policy continuity risk is assessed as medium to high. Pakistan has revised automotive and electric vehicle taxation in almost every budget cycle since the 2020 electric vehicle policy, exemptions have repeatedly been granted with end dates and then extended by a year at a time, and the two policies most relevant to this report were unapproved at the time of writing. A Chinese investor modelling a 5 to 10 year horizon should assume that concessional rates will be reviewed annually, that no incentive currently in force should be capitalised beyond its stated expiry of 30 June 2027, and that project economics should be tested against a scenario in which sales tax on locally manufactured electric vehicles rises toward the standard rate. Investments whose returns depend on a specific tariff differential persisting are materially riskier in Pakistan than in the comparator markets in section 15.

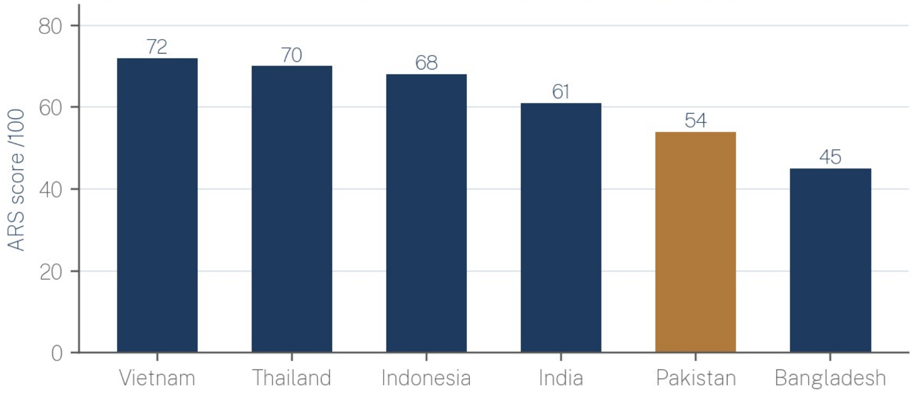
Foreign-exchange conditions have improved substantially and should now be treated as a manageable risk rather than a disqualifying one. Foreign investors repatriated 2.305 billion dollars in profits and dividends in fiscal year 2026, up from 2.219 billion dollars the previous year, with China the second-largest recipient at 486.5 million dollars. The State Bank cleared its repatriation backlog during fiscal year 2025 and reserves recovered from approximately 3 billion dollars in 2023 to a projected level near 17.8 billion dollars by June 2026. The residual exposure for EV investors is specific rather than general: business models that import cells, controllers and kits continuously while earning subsidised rupee revenue carry compounding exposure to both letter-of-credit processing and exchange-rate movement, and pack assembly for the domestic market is more exposed to this than vehicle import and distribution, which can be timed. ARS regards partial rupee-denominated financing and any credible export revenue stream as the two most effective hedges available in this market.

15. The Chinese Investor Landscape and the China+1 Comparison

Company	Local partner	Segment	Status	ARS note
BYD	Mega Motor Company (Hub Power and Mega Conglomerate)	Passenger EV and PHEV assembly	Operational imports since March 2025; Gharo plant in final commissioning	Approximately 150 million dollars; 25,000 units per year capacity; production expected Q3 or Q4 2026
Changan (Deepal)	Master Group (Master Changan Motors)	Passenger EV and REEV	Operational, expanding model range	Established JV manufacturing footprint predates EV push
Great Wall Motor	Sazgar Engineering	Passenger EV (ORA 03)	Launched	Sazgar also a major three-wheeler manufacturer
MG, Chery, JAC, FAW, BAIC, GAC	Various distributors and assemblers	Passenger EV, PHEV, REEV	Operational to announced	Model launch pipeline heavily electrified through 2026
Letin Auto	Local distributor	Small and micro EVs	Market entry announced	Low-cost segment entrant
Chaoniu Electric Motorcycle	Discussions via Board of Investment	Electric two-wheelers	Exploratory	Discussed phased manufacturing and technology transfer, January 2026
Yadea, Road Prince and others	Local assemblers	Electric scooters with swappable batteries	Operational	Closest existing model to the swapping opportunity
Juhang Energy Technology Group	Saigol Group	Lithium-ion batteries	Memorandum of understanding, 2026	Not yet a committed investment
Undisclosed Chinese battery firm	Punjab Board of Investment and Trade	Lithium battery plant, Faisalabad SEZ	Agreement signed, approximately 15 million dollars	Smallest announced project but closest to the identified opportunity

ARS compilation from public sources. Announcements, signed agreements and operational investments are distinguished in the status column and should not be treated as equivalent.

Figure 6 | ARS China+1 EV investment attractiveness score

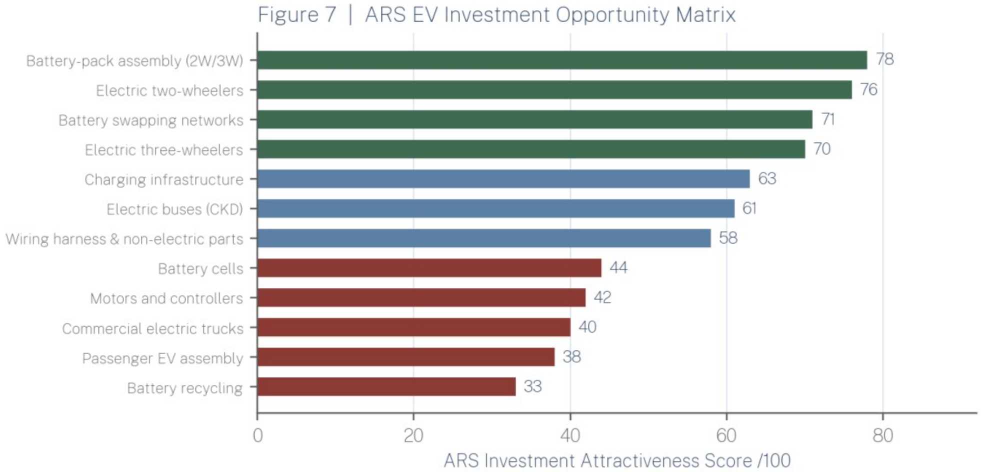


ARS composite. Weights: domestic market 20, policy stability 20, industrial base 15, FX and repatriation 15, cost base 10, export access 10, logistics 10. Scores are ARS judgements, not third-party index values.

Pakistan ranks fifth of six in the ARS China+1 comparison, ahead of Bangladesh and behind Vietnam, Thailand, Indonesia and India. It scores well on domestic market potential in the small-vehicle segment, on Chinese industrial incumbency, and on the breadth of its fiscal incentives. It scores poorly on industrial infrastructure, particularly electricity reliability and port logistics, on export platform value, since it has no equivalent of Thailand's or Vietnam's regional supply agreements, and on policy predictability. The practical conclusion is that Pakistan is a market to serve, not a base from which to export. An investor evaluating Pakistan against Indonesia or Vietnam for a regional manufacturing hub should choose the latter; an investor evaluating how to capture Pakistani demand should invest in Pakistan, and should do so in the small-vehicle and battery layers rather than in integrated vehicle manufacturing.

16. ARS Investment Opportunity Matrix, Watchlist and Methodology

Figure 7 | ARS EV Investment Opportunity Matrix



ARS composite of twelve weighted criteria set out in the methodology note. Green denotes invest now, blue denotes selective entry, red denotes wait.

The matrix scores twelve segments against twelve weighted criteria: market size and growth potential (20 points combined), import dependence and localisation potential (20), Chinese technological advantage (15), policy support (10), capital intensity, inverted (10), infrastructure requirement, inverted (10), competitive intensity, inverted (5), foreign-exchange risk, inverted (5), and regulatory risk and time to market (5). Scores are ARS analytical judgements informed by the evidence in this report and are not derived from a third-party index. The dashboard in section 2 uses the same underlying assessments aggregated at country level, with market opportunity, policy support and Chinese competitive advantage weighted at 20, 20 and 15 respectively, localisation potential and infrastructure readiness at 15 and 10, and the two inverted risk dimensions at 10 each.

Watch item	Why it matters	Signal to monitor
Auto Policy 2026-31	Sets tariff and localisation regime for 5 years	Cabinet approval and localisation schedule
National Lithium-Ion Battery Policy 2026-31	Determines pack assembly economics	National Tariff Board decision on component duties
Sales tax on locally made EVs	Currently 1%, expiring 30 June 2027	Finance Bill 2027 and interim notifications
PAVE Phase 3 allocation	Sets the volume floor for two and three-wheelers	Economic Coordination Committee approvals
Battery swapping standard	Determines whether a network can scale	NEECA or Ministry standardisation notification
BYD Gharo output	First real test of local EV assembly economics	Announced production volumes from Q4 2026
Provincial bus procurement	Largest committed commercial EV demand	Punjab PC-I releases and delivery schedules
Repatriation and reserves	Determines returns realisability	Monthly State Bank repatriation data

ARS Investor Watchlist. Each item is capable of materially changing the investment case set out in this report.

17. ARS Conclusion and Investment View

The evidence supports the central thesis, with one qualification. Pakistan's EV opportunity for Chinese investors is indeed larger in batteries, components, charging, commercial mobility and small vehicles than in passenger-vehicle assembly. The qualification is that the passenger-assembly segment is not merely less attractive but is already effectively occupied: BYD's vertically integrated position at Gharo, combined with the established Changan, Great Wall and MG joint ventures, means a new entrant in that segment would be competing for a market of a few thousand units a year against incumbents who control their own batteries and charging.

Chinese investors should invest now in battery-pack assembly for two and three-wheelers and stationary storage, in electric two and three-wheeler kit supply into the licensed assembler base, and in a captive financing capability attached to either. These are capital-light, address the demonstrated bottleneck, and are supported by legislated subsidy through at least fiscal year 2027. They should proceed selectively in battery swapping, where the prize is standard-setting and the risk is fragmentation, and in charging, where the viable position is anchored to a fleet or an existing retail footprint rather than merchant. They should wait on cell manufacture, motors, inverters and battery recycling until domestic volumes are an order of magnitude larger, and should avoid new capital in passenger EV assembly for now.

Three policy changes would materially improve the investment case: a mandated or industry-agreed swappable-battery standard, which would convert swapping from a set of captive pilots into an investable network; enactment of the lithium-ion battery policy with a multi-year, non-reversible tariff schedule for cells and pack components; and a credit guarantee or refinancing facility that removes the bank-approval bottleneck that has constrained subsidised two-wheeler deliveries. The three biggest risks are policy discontinuity around the 30 June 2027 expiry of current concessions, foreign-exchange exposure in continuous-import business models, and infrastructure reliability, particularly electricity supply to industrial and charging sites. The three most attractive opportunities over the next 3 to 5 years are battery-pack assembly, electric two-wheeler supply and financing, and swapping networks for commercial three-wheeler fleets.

ARS Investment View

Pakistan is a demand market, not an export platform. It merits targeted, capital-light investment in the battery and small-vehicle layers of the value chain, where Chinese technological advantage is greatest, import dependence is near total, minimum efficient scale is low and public subsidy is legislated and funded. It does not currently merit capital-intensive integrated manufacturing, and any investment thesis that depends on a specific tariff differential surviving beyond 30 June 2027 should be stress-tested before commitment. ARS composite attractiveness: 61 out of 100, selective entry.

18. Sources, Data Standards and Methodology

This report draws on Government of Pakistan sources including the Ministry of Industries and Production, the Engineering Development Board, the Finance Bill 2026 and associated budget documentation, the Economic Survey for fiscal year 2025-26, the National Electric Power Regulatory Authority, the National Energy Efficiency and Conservation Authority, the State Bank of Pakistan and the provincial governments of Punjab, Sindh and Khyber Pakhtunkhwa. Industry data is drawn from Arif Habib Limited and specialist motorcycle market data providers. Company information is drawn from public announcements by BYD, Mega Motor Company, Hub Power, Master Changan Motors, Sazgar Engineering and battery manufacturers including EV Technologies, Crown Group and Zilo Energy, together with reporting by Reuters, Bloomberg, Nikkei Asia, Dawn, Business Recorder, The Express Tribune, Profit and Arab News.

Three data-integrity qualifications apply throughout. First, vehicles manufactured, imported, registered and operating are reported inconsistently in Pakistani sources and are treated separately here rather than aggregated. Second, official battery demand estimates for 2030 differ by an order of magnitude between the 8.7 GWh projection and the 65 to 100 gigawatt figure cited by the Engineering Development Board, and ARS has adopted the former. Third, reporting on completely built-up EV import duty for fiscal year 2026-27 is contradictory, and ARS has presented both accounts rather than selecting between them. All scores, matrices, growth calculations and rankings identified as ARS are analytical products of Ardent Research Solutions, derived from the evidence cited, and represent professional judgement rather than measured data.

This report is standalone market intelligence. Questions it raises but does not answer, including which Pakistani firms are suitable joint-venture partners, which special economic zone best combines power reliability and logistics for pack assembly, what a specific charging or swapping deployment would require in a named city, and how a defined project would model against Vietnamese or Indonesian alternatives, can be addressed through commissioned research. Ardent Research Solutions provides bespoke investment intelligence to institutional and corporate clients evaluating entry into Pakistani markets.

About the Author

Reza Khan is a researcher with an M.Phil. in Government and Public Policy from the National Defence University, Islamabad, and an MBA in Finance. His professional background spans business strategy, international trade, economics and organizational leadership. His research focuses on public policy, governance, strategic communication, and the relationship between technology, institutions and society.

